Irri Nissy

am25mtech11003@iith.ac.in | irrinissy321@gmail.com | +91 8125698869 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.17
Sumathi Devi Junior College for girls	Class XII (Telangana State Board)	2015	82.10%
St. Ann's High School	Class X (Andhra Pradesh State Board)	2013	8.70%

PROJECTS

Microstructural Characterisation of EB-PBF Processed MAR-M247 Superalloy (M.Tech Thesis)

- Investigating microstructural evolution and heat-treatment effects in EB-PBF processed MAR-M247 superalloy.
- Characterising microstructure using OM, SEM, EBSD and TEM to correlate phase evolution with material behaviour.
- Evaluating mechanical performance through high-temperature tensile testing, stress-rupture, hardness and fractographic analysis.

Finite Element Analysis of Corneal Stress in Healthy and Keratoconus Eyes

- Developed a 3D finite element model of the human cornea under intraocular pressure.
- Implemented a Mooney-Rivlin hyperelastic material model to simulate corneal behaviour.
- Conducted nonlinear simulations in ABAQUS to compare healthy and keratoconus conditions

Hydrogen-Air Mixing in a 2D Scramjet Combustor

- Modelled hydrogen-air flow in ANSYS Fluent for a scramjet combustor with a wedge-type strut injector.
- Investigated strut-angle effects on shock structure, wake development, and hydrogen-air mixing using flow contours and mass-fraction profiles.
- Optimised strut geometry and lateral placement to enhance hydrogen-air mixing and improve combustor performance.

Design and Additive Manufacturing of a Pawl and Ratchet Mechanism

- Designed a functional pawl-ratchet mechanism using parametric CAD in Autodesk Inventor and applied DfAM principles for improved printability.
- Fabricated and evaluated the prototype using FDM (Creality Ender 3 V2, PLA).

INTERNSHIPS

TRANSCO, Trainee

- Executed assigned site activities under the guidance of the Assistant Engineer.
- Coordinated work execution to meet prescribed timelines and quality requirements.

South Central Railway, Intern

- Gained practical exposure to diesel locomotive power-pack maintenance, inspection, and operational procedures at South Central Railway.

SKILLS

Simulation and Analysis: FEM, CFD

Engineering Software: ABAQUS, ANSYS Fluent, MATLAB, Autodesk Inventor

Additive Manufacturing: EB-PBF, SLM, Ultimaker Cura, FDM 3D Printing

Microstructural Characterisation: SEM, OM, EBSD, TEM

RELEVANT COURSES

- Finite Element Methods (FEM)
- Computational Fluid Dynamics (CFD) Tools
- Machine Learning and its Applications
- Design for Metal Additive Manufacturing
- Product Design and Prototyping
- Metallurgy of Welding and Additive Manufacturing
- Mechanical Behaviour of Materials
- Design for Manufacturing and Assembly

EXTRACURRICULAR

Certifications

- **Robotics Workshop - IIT Hyderabad, WAC 2019**

Participated in a Robotics Workshop conducted by Kyrion Technologies Pvt. Ltd. at IIT Hyderabad.

Volunteering

- **VOICE 4 Girls,** Counsellor

Mentored and guided school students through interactive counselling and engagement activities.

Hobbies and Interests

- Swimming | Running | Exploring diverse global music | Anime